CODE REPORT

BlazeDemo Automation Framework

Project Name: BlazeDemo Flight Booking Automation Framework

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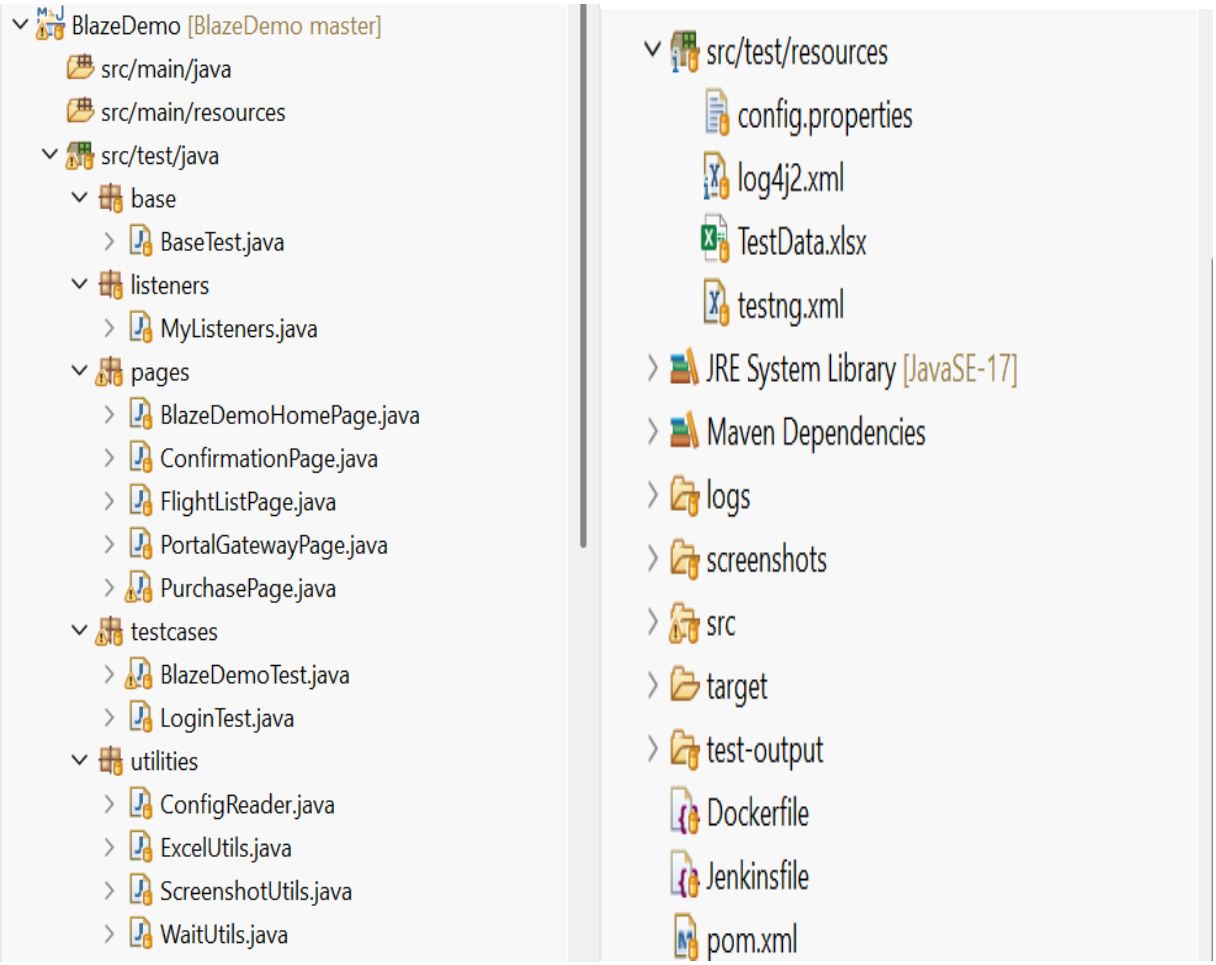
Github: https://github.com/AYUSH-YL/Capstone_project

Site: https://blazedemo.com/reserve.php

Project Structure:



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CODE:

BaseTest.java:

```
package base;


import org.apache.logging.log4j.LogManager;
import org.apache.logging.log4j.Logger;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.testng.annotations.AfterClass;
import org.testng.annotations.BeforeClass;
import utilities.ScreenshotUtils;

public class BaseTest {

    protected static final Logger log = LogManager.getLogger(BaseTest.class);
    protected WebDriver driver;

    @BeforeClass(alwaysRun = true)
    public void setUp() {
        log.info("Initializing browser automation setup.");

        ChromeOptions options = new ChromeOptions();
        options.addArguments("--remote-allow-origins=*");
        options.addArguments("--no-sandbox");
        options.addArguments("--disable-dev-shm-usage");
```

```
        if (System.getenv("RUNNING_IN_DOCKER") != null) {
            log.info("System environment variable detected: Docker Pipeline. Forcing headless configuration flags.");
            options.addArguments("--headless=new");
            options.addArguments("--disable-gpu");
            options.addArguments("--window-size=1920,1080");
            options.setBinary("/usr/bin/google-chrome");
        } else {
            log.info("System environment variable missing: Local Eclipse Sandbox. Launching visible GUI browser panel.");
        }

        driver = new ChromeDriver(options);
        driver.manage().window().maximize();

        log.info("Chrome Browser instance initialized successfully.");
    }
    @AfterClass(alwaysRun = true)
    public void tearDown() {
        if (driver != null) {
            driver.quit();
            log.info("Browser instance terminated cleanly.");
        }
    }

    public WebDriver getDriver() {
        return this.driver;
    }
```

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```
    public void takeCheckpointScreenshot(String stepName) {
        log.info("Triggering manual checkpoint screenshot: " + stepName);
        ScreenshotUtils.captureScreenshot(driver, stepName);
    }
}
```

MyLiateners.java:

```
package listeners;

import com.aventstack.extentreports.ExtentReports;
import com.aventstack.extentreports.ExtentTest;
import com.aventstack.extentreports.Status;
import com.aventstack.extentreports.reporter.ExtentSparkReporter;
import com.aventstack.extentreports.reporter.configuration.Theme;
import org.testng.ITestContext;
import org.testng.ITestListener;
import org.testng.ITestResult;
import base.BaseTest;

public class MyListeners implements ITestListener {

    private static ExtentReports extent;
    private static ThreadLocal<ExtentTest> testThread = new ThreadLocal<>();

    @Override
    public void onStart(ITestContext context) {
```

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```
        ExtentSparkReporter sparkReporter = new
ExtentSparkReporter("target/ExtentReport.html");
        sparkReporter.config().setDocumentTitle("BlazeDemo Automation Report");
        sparkReporter.config().setReportName("End To End Regression Results");
        sparkReporter.config().setTheme(Theme.DARK);

        extent = new ExtentReports();
        extent.attachReporter(sparkReporter);
        extent.setSystemInfo("Environment", "QA Pipeline");
        extent.setSystemInfo("Author", "Ayush");
        extent.setSystemInfo("OS", System.getProperty("os.name"));
    }

    @Override
    public void onTestStart(ITestResult result) {
        System.out.println("Starting Test Execution: " + result.getName());
        ExtentTest test = extent.createTest(result.getName());
        testThread.set(test);
    }

    @Override
    public void onTestSuccess(ITestResult result) {
        System.out.println("TEST PASSED: " + result.getName());
        testThread.get().log(Status.PASS, "Test Case Passed: " + result.getName());
    }

    @Override
```

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```
    public void onTestFailure(ITestResult result) {
        System.out.println("TEST FAILED: " + result.getName() + " -> Taking automated screenshot.");
        testThread.get().log(Status.FAIL, "Test Case Failed: " + result.getName());
        testThread.get().log(Status.FAIL, result.getThrowable());

        try {
            Object testClass = result.getInstance();
            org.openqa.selenium.WebDriver driver = ((BaseTest) testClass).getDriver();

            if (driver != null) {
                String screenshotPath = utilities.ScreenshotUtils.captureScreenshot(driver, result.getName());
                testThread.get().addScreenCaptureFromPath("../" + screenshotPath);
            }
        } catch (Exception e) {
            System.out.println("Exception thrown while attaching screenshot to Extent Report: " + e.getMessage());
        }
    }

    @Override
    public void onTestSkipped(ITestResult result) {
        System.out.println("Test Skipped: " + result.getName());
        testThread.get().log(Status.SKIP, "Test Case Skipped: " + result.getName());
    }

    @Override
```

```
    public void onFinish(ITestContext context) {
        if (extent != null) {
            extent.flush();
        }
    }
    public static ExtentTest getTestLog() {
        return testThread.get();
    }
}
```

BlazeDemoHomePage.java:

```
package pages;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.support.ui.Select;
import utilities.WaitUtils;

public class BlazeDemoHomePage {
    private WebDriver driver;

    private By departureDropdown = By.name("fromPort");
    private By destinationDropdown = By.name("toPort");
    private By findFlightsButton = By.cssSelector("input[type='submit'].btn-primary");
    private By homeButtonLink = By.linkText("home");

    public BlazeDemoHomePage(WebDriver driver) {
```

```
        this.driver = driver;
    }

    public void clickHomeButtonLink() {
        WaitUtils.waitForElementClickable(driver, homeButtonLink, 10).click();
    }

    public void selectDepartureCity(String departure) {
        WaitUtils.waitForElementVisible(driver, departureDropdown, 10);
        Select select = new Select(driver.findElement(departureDropdown));
        select.selectByValue(departure);
    }

    public void selectDestinationCity(String destination) {
        Select select = new Select(driver.findElement(destinationDropdown));
        select.selectByValue(destination);
    }

    public void clickFindFlights() {
        driver.findElement(findFlightsButton).click();
    }
}
```

ConfirmationPage.java:

```
package pages;

import org.openqa.selenium.By;
```

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```
import org.openqa.selenium.WebDriver;

import utilities.WaitUtils;


public class ConfirmationPage {

   private WebDriver driver;


   private By thankYouHeader = By.xpath("//h1[contains(text(),'Thank you for your purchase
today!')]");


   public ConfirmationPage(WebDriver driver) {

      this.driver = driver;

   }


   public String getConfirmationMessage() {

      return WaitUtils.waitForElementVisible(driver, thankYouHeader, 10).getText().trim();

   }

}
```

FlightList.java:

```
package pages;


import org.openqa.selenium.By;

import org.openqa.selenium.WebDriver;

import utilities.WaitUtils;


public class FlightListPage {

   private WebDriver driver;
```

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```
    private By flightsHeader = By.xpath("//h3[contains(text(),'Flights from')]");
    private By chooseFlightButton = By.cssSelector("input[type='submit'][value='Choose This Flight']");

    public FlightListPage(WebDriver driver) {
        this.driver = driver;
    }

    public boolean isFlightListDisplayed() {
        return WaitUtils.waitForElementVisible(driver, flightsHeader, 10).isDisplayed();
    }

    public void chooseFirstAvailableFlight() {
        WaitUtils.waitForElementClickable(driver, chooseFlightButton, 10).click();
    }
}
```

PortalGatewayPage.java:

```
package pages;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import utilities.WaitUtils;

public class PortalGatewayPage {
    private WebDriver driver;

    private By loginEmailField = By.id("email");
```

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```
    private By loginPasswordField = By.id("password");
    private By loginSubmitButton = By.cssSelector("button[type='submit']");

    private By registerLinkButton = By.linkText("Register");

    private By regNameField = By.id("name");
    private By regCompanyField = By.id("company");
    private By regEmailField = By.id("email");
    private By regPasswordField = By.id("password");
    private By regConfirmPasswordField = By.id("password-confirm");
    private By regSubmitButton = By.cssSelector("button[type='submit']");

    public PortalGatewayPage(WebDriver driver) {
        this.driver = driver;
    }

    public void clickRegisterLink() {
        WaitUtils.waitForElementClickable(driver, registerLinkButton, 10).click();
    }

    public void executeNegativeRegistration(String name, String company, String email, String
password) {
        WaitUtils.waitForElementVisible(driver, regNameField, 10);
        driver.findElement(regNameField).sendKeys(name);
        driver.findElement(regCompanyField).sendKeys(company);
        driver.findElement(regEmailField).sendKeys(email);
        driver.findElement(regPasswordField).sendKeys(password);
```

```
        driver.findElement(regConfirmPasswordField).sendKeys(password);
        driver.findElement(regSubmitButton).click();
    }

    public void executeNegativeLogin(String email, String password) {
        WaitUtils.waitForElementVisible(driver, loginEmailField, 10);
        driver.findElement(loginEmailField).sendKeys(email);
        driver.findElement(loginPasswordField).sendKeys(password);
        driver.findElement(loginSubmitButton).click();
    }

    public boolean verify419ErrorPageState() {
        String source = driver.getPageSource();
        return source.contains("419") || source.contains("Page Expired");
    }

    public void navigateBackTwoSteps() {
        driver.navigate().back();
        driver.navigate().back();
    }
}
```

PurchasePage.java:

```
package pages;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
```

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```
import org.openqa.selenium.support.ui.Select;
import utilities.WaitUtils;
import org.openqa.selenium.support.ui.Select;
import org.openqa.selenium.NoSuchElementException;
import org.apache.logging.log4j.LogManager;
import org.apache.logging.log4j.Logger;

public class PurchasePage {
        private static final Logger log = LogManager.getLogger(PurchasePage.class);
    private WebDriver driver;

    private By purchaseHeader = By.xpath("//h2[contains(text(),'Your flight')]");
    private By totalCostText = By.xpath("//p[contains(text(),'Total Cost:')]/em");

    private By inputName = By.id("inputName");
    private By addressField = By.id("address");
    private By cityField = By.id("city");
    private By stateField = By.id("state");
    private By zipCodeField = By.id("zipCode");
    private By cardTypeDropdown = By.id("cardType");
    private By creditCardNumberField = By.id("creditCardNumber");
    private By nameOnCardField = By.id("nameOnCard");
    private By purchaseFlightButton = By.cssSelector("input[type='submit'][value='Purchase
Flight']");

    public PurchasePage(WebDriver driver) {
        this.driver = driver;
```

```
    }

    public boolean isPurchasePageDisplayed() {
        return WaitUtils.waitForElementVisible(driver, purchaseHeader, 10).isDisplayed();
    }

    public String getTotalCost() {
        return driver.findElement(totalCostText).getText().trim();
    }

    public void fillPersonalAndPaymentDetails(String name, String address, String city, String
state, String zip, String cardType, String cardNumber, String cardName) {
        driver.findElement(inputName).sendKeys(name);
        driver.findElement(addressField).sendKeys(address);
        driver.findElement(cityField).sendKeys(city);
        driver.findElement(stateField).sendKeys(state);
        driver.findElement(zipCodeField).sendKeys(zip);

        Select selectCard = new Select(driver.findElement(cardTypeDropdown));
        try {
            selectCard.selectByValue(cardType.toLowerCase());
        } catch (NoSuchElementException e) {
            log.warn("Card type '" + cardType + "' not found. Defaulting to 'Visa'.");
            selectCard.selectByVisibleText("Visa");
        }

        driver.findElement(creditCardNumberField).sendKeys(cardNumber);
```

```
        driver.findElement(nameOnCardField).sendKeys(cardName);
    }

    public void clickPurchaseFlight() {
        driver.findElement(purchaseFlightButton).click();
    }
}
```

BlazedemoTest.java:

```
package testcases;

import org.testng.Assert;
import org.testng.annotations.DataProvider;
import org.testng.annotations.Factory;
import org.testng.annotations.Test;

import base.BaseTest;
import pages.BlazeDemoHomePage;
import pages.ConfirmationPage;
import pages.FlightListPage;
import pages.PurchasePage;
import utilities.ExcelUtils;

public class BlazeDemoTest extends BaseTest {

    private BlazeDemoHomePage homePage;
    private FlightListPage flightListPage;
```

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```
private PurchasePage purchasePage;
private ConfirmationPage confirmationPage;

private final String departure;
private final String destination;
private final String name;
private final String address;
private final String city;
private final String state;
private final String zip;
private final String cardType;
private final String cardNumber;
private final String cardName;

@DataProvider(name = "bookingData")
public static Object[][] bookingData() {
   return ExcelUtils.getTestData("src/test/resources/TestData.xlsx", "BookingSheet");
}

@Factory(dataProvider = "bookingData")
public BlazeDemoTest(String departure, String destination, String name, String address,
                     String city, String state, String zip, String cardType,
                     String cardNumber, String cardName) {
   this.departure = departure;
   this.destination = destination;
   this.name = name;
   this.address = address;
```

```
        this.city = city;
        this.state = state;
        this.zip = zip;
        this.cardType = cardType;
        this.cardNumber = cardNumber;
        this.cardName = cardName;
    }

    @Test(priority = 1, groups = { "end-to-end" })
    public void step2_searchFlights() {
        log.info("Selecting Route -> " + departure + " to " + destination);

        driver.get(utilities.ConfigReader.getUrl());

        homePage = new BlazeDemoHomePage(driver);
        Assert.assertEquals(driver.getTitle(), "BlazeDemo", "Page title mismatch on Home Page");
        Assert.assertEquals(driver.getCurrentUrl(), "https://blazedemo.com/index.php", "Home
Page URL mismatch");

        homePage.selectDepartureCity(departure);
        homePage.selectDestinationCity(destination);
        homePage.clickFindFlights();
    }

    @Test(priority = 2, dependsOnMethods = { "step2_searchFlights" }, groups = { "end-to-end"
})
    public void step3_validateAndSelectFlight() {
        log.info("Verifying flight grid selection table.");
```

```
        flightListPage = new FlightListPage(driver);
        Assert.assertTrue(flightListPage.isFlightListDisplayed(),
                "Validation Failure: Flights listing table is missing.");
        flightListPage.chooseFirstAvailableFlight();
    }

    @Test(priority = 3, dependsOnMethods = { "step3_validateAndSelectFlight" }, groups = { "end-to-end" })
    public void step4_fillPassengerFormAndBook() {
        log.info("Filling passenger details for: " + name);

        purchasePage = new PurchasePage(driver);
        Assert.assertTrue(purchasePage.isPurchasePageDisplayed(),
                "Validation Failure: Purchase page header not displayed.");

        String cleanName = name.replaceAll("[^a-zA-Z0-9]", "");

        purchasePage.fillPersonalAndPaymentDetails(name, address, city, state, zip,
                cardType, cardNumber, cardName);

        takeCheckpointScreenshot("FormInputCheckpoint_Passenger_" + cleanName);
        purchasePage.clickPurchaseFlight();
    }

    @Test(priority = 4, dependsOnMethods = { "step4_fillPassengerFormAndBook" }, groups = { "end-to-end" })
    public void step5_validateConfirmationReceipt() {
        log.info("Asserting completion receipts for passenger: " + name);
```

```
        confirmationPage = new ConfirmationPage(driver);
        String confirmationMessage = confirmationPage.getConfirmationMessage();

        Assert.assertEquals(confirmationMessage, "Thank you for your purchase today!",
                "Validation Failure: Success banner mismatch!");

        String cleanName = name.replaceAll("[^a-zA-Z0-9]", "");
        takeCheckpointScreenshot("FinalBookingConfirmationReceipt_" + cleanName);
    }
}
```

LoginTest.java:

```
package testcases;

import base.BaseTest;
import org.testng.Assert;
import org.testng.annotations.Test;
import pages.BlazeDemoHomePage;
import pages.PortalGatewayPage;

public class LoginTest extends BaseTest {

    private BlazeDemoHomePage homePage;
    private PortalGatewayPage portalPage;

    @Test(priority = 1, groups = { "end-to-end" })
```

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```
    public void validateBrokenPortalGateways() {
        log.info("Starting Portal Security Regression Sweep (Once).");

        String appUrl = utilities.ConfigReader.getUrl();
        log.info("Navigating directly to application index: " + appUrl);
        driver.get(appUrl);

        homePage = new BlazeDemoHomePage(driver);
        portalPage = new PortalGatewayPage(driver);

        homePage.clickHomeButtonLink();
        portalPage.clickRegisterLink();

        log.info("Injecting hardcoded registration placeholders.");
        portalPage.executeNegativeRegistration("DemoUser", "Wipro", "test@wipro.com",
"Password123");

        takeCheckpointScreenshot("Registration_419_Error_Page");
        Assert.assertTrue(portalPage.verify419ErrorPageState(), "Error: Registration failed to
trigger a 419 Page state.");
        log.info("Known error : Registration 419 Page.");

        portalPage.navigateBackTwoSteps();

        log.info("Injecting hardcoded login placeholders.");
        portalPage.executeNegativeLogin("login@wipro.com", "SecurePass99");

        takeCheckpointScreenshot("Login_419_Error_Page");
```

```
        Assert.assertTrue(portalPage.verify419ErrorPageState(), "Error: Login failed to trigger a
419 Page state.");
        log.info("Known error : 419 Page.");

        portalPage.navigateBackTwoSteps();
        log.info("Portal Sweep Complete.");
    }
}
```

ConfigReader.java:

```
package utilities;

import java.io.FileInputStream;
import java.io.IOException;
import java.util.Properties;

public class ConfigReader {

    private static Properties properties;
    private static final String PROPERTY_FILE_PATH = "src/test/resources/config.properties";

    static {
        try (FileInputStream fileInputStream = new FileInputStream(PROPERTY_FILE_PATH)) {
            properties = new Properties();
            properties.load(fileInputStream);
        } catch (IOException e) {
            System.out.println("Critical Exception: Failed to load environment properties
configuration mapping file at " + PROPERTY_FILE_PATH);
```

```
            e.printStackTrace();
            throw new RuntimeException("Configuration file initialization aborted due to critical I/O stream failure.");
        }
    }

    public static String getProperty(String key) {
        String value = properties.getProperty(key);
        if (value == null) {
            System.out.println("Warning: Requested configuration mapping key '" + key + "' was not found inside your properties structure.");
        }
        return value;
    }

    public static String getUrl() {
        return getProperty("url");
    }

    public static String getBrowser() {
        return getProperty("browser");
    }

    public static int getExplicitWaitTimeout() {
        return Integer.parseInt(getProperty("explicitWait"));
    }
}
```

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ExcelUtils.java:

```
package utilities;


import java.io.FileInputStream;

import java.io.IOException;

import org.apache.poi.ss.usermodel.DataFormatter;

import org.apache.poi.xssf.usermodel.XSSFCell;

import org.apache.poi.xssf.usermodel.XSSFRow;

import org.apache.poi.xssf.usermodel.XSSFSheet;

import org.apache.poi.xssf.usermodel.XSSFWorkbook;


public class ExcelUtils {


    public static Object[][] getTestData(String filePath, String sheetName) {
        XSSFWorkbook workbook = null;
        FileInputStream fileInputStream = null;
        Object[][] data = null;


        try {
            fileInputStream = new FileInputStream(filePath);
            workbook = new XSSFWorkbook(fileInputStream);
            XSSFSheet sheet = workbook.getSheet(sheetName);


            int totalRows = sheet.getLastRowNum();
            XSSFRow headerRow = sheet.getRow(0);
            int totalColumns = headerRow.getLastCellNum();
```

```
            data = new Object[totalRows][totalColumns];
            DataFormatter formatter = new DataFormatter();
            for (int i = 1; i <= totalRows; i++) {
                XSSFRow row = sheet.getRow(i);
                for (int j = 0; j < totalColumns; j++) {
                    if (row == null) {
                        data[i - 1][j] = "";
                    } else {
                        XSSFCell cell = row.getCell(j);
                        data[i - 1][j] = formatter.formatCellValue(cell);
                    }
                }
            }
        } catch (IOException e) {
            System.out.println("Critical Exception: Failed to read from Excel data matrix at path " +
filePath);
            e.printStackTrace();
        } finally {
            try {
                if (workbook != null) workbook.close();
                if (fileInputStream != null) fileInputStream.close();
            } catch (IOException e) {
                System.out.println("Exception encountered while closing Excel streams: " +
e.getMessage());
            }
        }
        return data;
    }
```

```
}
```

ScreenshotUtils.java:

```
package utilities;


import java.io.File;

import java.io.IOException;

import java.text.SimpleDateFormat;

import java.util.Date;

import org.openqa.selenium.OutputType;

import org.openqa.selenium.TakesScreenshot;

import org.openqa.selenium.WebDriver;

import org.openqa.selenium.io.FileHandler;


public class ScreenshotUtils {


    public static String captureScreenshot(WebDriver driver, String screenshotName) {

        String timestamp = new SimpleDateFormat("yyyyMMdd_HHmmss").format(new Date());

        String fileName = screenshotName + "_" + timestamp + ".png";


        File directory = new File("screenshots");

        if (!directory.exists()) {

            directory.mkdir();

        }


        File source = ((TakesScreenshot) driver).getScreenshotAs(OutputType.FILE);

        String destinationPath = "screenshots/" + fileName;
```

```
        File destination = new File(destinationPath);
        try {
            FileHandler.copy(source, destination);
            System.out.println("Screenshot captured successfully: " + destinationPath);
        } catch (IOException e) {
            System.out.println("Failed to capture screenshot: " + e.getMessage());
        }
        return destinationPath;
    }
}
```

WaitUtils.java:

```
package utilities;

import java.time.Duration;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;

public class WaitUtils {

    public static WebElement waitForElementVisible(WebDriver driver, By locator, int timeoutInSeconds) {
        WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(timeoutInSeconds));
        return wait.until(ExpectedConditions.visibilityOfElementLocated(locator));
```

```
    }

    public static WebElement waitForElementClickable(WebDriver driver, By locator, int
timeoutInSeconds) {
        WebDriverWait wait = new WebDriverWait(driver,
Duration.ofSeconds(timeoutInSeconds));
        return wait.until(ExpectedConditions.elementToBeClickable(locator));
    }
}
```

Config.properties:

```
url=https://blazedemo.com/index.php
browser=chrome

implicitWait=10
explicitWait=10

excelTestDataPath=src/test/resources/TestData.xlsx
```

log4j2.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<Configuration status="WARN">
    <Properties>
        <Property name="baseDir">logs</Property>
    </Properties>

    <Appenders>
```

```
        <Console name="ConsoleAppender" target="SYSTEM_OUT">
            <PatternLayout pattern="%d{yyyy-MM-dd HH:mm:ss} [%t] %-5level %logger{36} -
%msg%n"/>
        </Console>

        <RollingFile name="FileAppender" fileName="${baseDir}/automation.log"
                     filePattern="${baseDir}/automation-%d{yyyy-MM-dd}-%i.log.gz">
            <PatternLayout pattern="%d{yyyy-MM-dd HH:mm:ss} [%t] %-5level %logger{36} -
%msg%n"/>
            <Policies>
                <SizeBasedTriggeringPolicy size="10 MB" />
            </Policies>
            <DefaultRolloverStrategy max="5"/>
        </RollingFile>
    </Appenders>

    <Loggers>
        <Root level="INFO">
            <AppenderRef ref="ConsoleAppender"/>
            <AppenderRef ref="FileAppender"/>
        </Root>
    </Loggers>
</Configuration>
```

TestData.xlxs:

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departure	destination	name	address	city	state	zip	cardType	cardNumber	cardName
Paris	London	John##Doe!!	123 Tech Lane	Noida	UP	201301	Visa	FOUR_ONE_ONE_ONE	John Doe
Portland	Rome	Amit_Sharma_$$	456 IT Hub	Bangalore	Karnataka	560001	Mastercard	XYZ-INVALID-CARD	Amit Sharma
Boston	Berlin	<Admin>	789 Web Drive	Mumbai	MH	400001	American Express	0000AAAA0000	Admin

Testng.xml:

```
<?xml version="1.0" encoding="UTF-8"?>

<!DOCTYPE suite SYSTEM "https://testng.org/testng-1.0.dtd">

<suite name="BlazeDemo Booking Suite">

    <listeners>

        <listener class-name="listeners.MyListeners" />

    </listeners>


    <test name="End To End Flight Booking Test">

        <groups>

            <run>

                <include name="end-to-end" />

            </run>

        </groups>

        <classes>

            <class name="testcases.LoginTest" />


            <class name="testcases.BlazeDemoTest" />

        </classes>

    </test>

</suite>
```

CODE REPORT

Pom.xml:

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>BlazeDemo</groupId>
  <artifactId>BlazeDemo</artifactId>
  <version>0.0.1-SNAPSHOT</version>

  <properties>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  </properties>

  <dependencies>
    <dependency>
        <groupId>org.seleniumhq.selenium</groupId>
        <artifactId>selenium-java</artifactId>
        <version>4.20.0</version>
    </dependency>

    <dependency>
        <groupId>org.testng</groupId>
        <artifactId>testng</artifactId>
        <version>7.9.0</version>
    </dependency>

    <dependency>
```

```
    <groupId>org.apache.poi</groupId>
    <artifactId>poi-ooxml</artifactId>
    <version>5.2.5</version>
  </dependency>

  <dependency>
    <groupId>org.apache.logging.log4j</groupId>
    <artifactId>log4j-api</artifactId>
    <version>2.23.1</version>
  </dependency>

  <dependency>
    <groupId>org.apache.logging.log4j</groupId>
    <artifactId>log4j-core</artifactId>
    <version>2.23.1</version>
  </dependency>

  <dependency>
    <groupId>com.aventstack</groupId>
    <artifactId>extentreports</artifactId>
    <version>5.1.1</version>
  </dependency>
</dependencies>

<build>
  <plugins>
    <plugin>
```

CODE REPORT

```
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-compiler-plugin</artifactId>
      <version>3.13.0</version>
      <configuration>
        <source>17</source>
        <target>17</target>
      </configuration>
    </plugin>

    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-surefire-plugin</artifactId>
      <version>3.5.3</version>
      <configuration>
        <suiteXmlFiles>
          <suiteXmlFile>src/test/resources/testng.xml</suiteXmlFile>
        </suiteXmlFiles>
      </configuration>
    </plugin>
  </plugins>
 </build>
</project>
```

Jenkinsfile:

```
pipeline {
    agent any
```

CODE REPORT

```
    stages {
        stage('Docker Environment Verification') {
            steps {
                echo 'Verifying that Jenkins has access to the local Docker engine.'
                bat 'docker --version'
            }
        }

        stage('Docker Build Lifecycle') {
            steps {
                echo 'Compiling image artifact layer from project Dockerfile specifications.'
                bat 'docker build -t testng-framework .'
            }
        }

        stage('Docker Container Regression Run') {
            steps {
                echo 'Spanning up sandboxed container node to launch TestNG framework headless
suite.'
                bat 'docker run --name blazedemo-run testng-framework'
            }
        }
        stage('Docker Container Regression Run') {
            steps {
                echo 'Clearing any stale containers from prior failed pipeline runs.'
                bat 'docker rm -f blazedemo-run || ver>nul'
```

CODE REPORT

```
            echo 'Spanning up sandboxed container node to launch TestNG framework headless suite.'
            bat 'docker run --name blazedemo-run testng-framework'
        }
    }
}

post {
    always {
        echo 'Extracting isolated test results and screenshots from inside the closed container instance.'
        bat 'docker cp blazedemo-run:/app/screenshots ./screenshots'
        bat 'docker cp blazedemo-run:/app/target/ExtentReport.html ./target/ExtentReport.html'
        bat 'docker cp blazedemo-run:/app/target/surefire-reports ./target/surefire-reports'

        echo 'Cleaning up execution workspace nodes.'
        bat 'docker rm blazedemo-run'

        junit '**/target/surefire-reports/*.xml'
        archiveArtifacts artifacts: 'screenshots/**/*', allowEmptyArchive: true
        archiveArtifacts artifacts: 'target/ExtentReport.html', allowEmptyArchive: true
    }
    success {
        echo 'Success: Containerized pipeline execution passed completely!'
    }
    failure {
        echo 'Error: Regression failure detected during isolated execution.'
    }
```

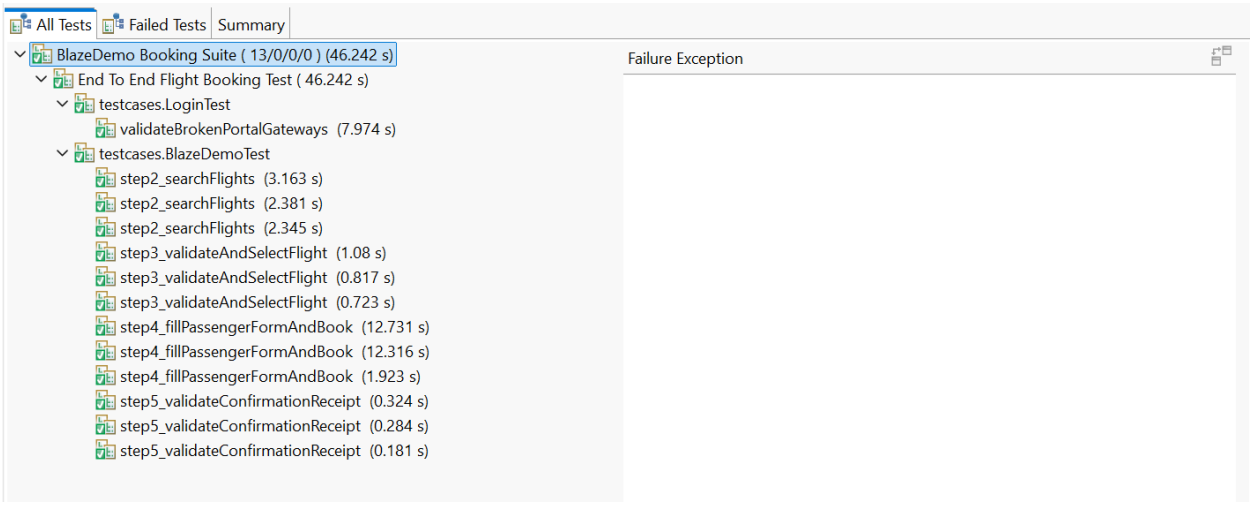
```
    }
}
```

Dockerfile:

```
FROM maven:3.8.8-eclipse-temurin-17

RUN apt-get update && apt-get install -y wget gnupg2 \
    && wget -q -O - https://dl-ssl.google.com/linux/linux_signing_key.pub | apt-key add - \
    && echo "deb [arch=amd64] http://dl.google.com/linux/chrome/deb/ stable main" >> /etc/apt/sources.list.d/google-chrome.list \
    && apt-get update && apt-get install -y google-chrome-stable \
    && rm -rf /var/lib/apt/lists/*

WORKDIR /app

COPY pom.xml .

RUN mvn dependency:go-offline -B

COPY src ./src

ENV RUNNING_IN_DOCKER=true

CMD ["mvn", "clean", "test"]
```

CODE REPORT

Code Excecution Result:




```
TEST PASSED: step5_validateConfirmationReceipt
2026-06-16 14:51:09 [main] INFO  base.BaseTest - Browser instance terminated cleanly.
Starting Test Execution: step5_validateConfirmationReceipt
2026-06-16 14:51:09 [main] INFO  base.BaseTest - Asserting completion receipts for passenger: John##Doe!!
2026-06-16 14:51:09 [main] INFO  base.BaseTest - Triggering manual checkpoint screenshot: FinalBookingConfirmationReceipt_John
Screenshot captured successfully: screenshots/FinalBookingConfirmationReceipt_JohnDoe_20260616_145109.png
TEST PASSED: step5_validateConfirmationReceipt
2026-06-16 14:51:10 [main] INFO  base.BaseTest - Browser instance terminated cleanly.

================================================
BlazeDemo Booking Suite
Total tests run: 13, Passes: 13, Failures: 0, Skips: 0
================================================
```